

Case Study: *Data-driven Engineering (DDE)* Course

AUTO EME OPTION

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CS1: Machine learning-based torque and SoC estimation of an e-bike

Task description:

- Retrieve the data automatically from FIT files, i.e., power (rider and motor), distance, speed, cadence, altitude and Vehicle speed;
- Plot and analyze the data from real driving scenarios, incl. torque of the rider and motor;
- Develop an equivalent circuit model (ECM) of the battery, and physics-based model of the e-bike.
- Generalize model parameters from scalar values to structured representations (e.g., multi-dimensional look-up tables).
- Develop and compare a data-driven or hybrid (physics-informed + data-driven) model to improve torque and SoC prediction accuracy.

➤ Aim of this case study:

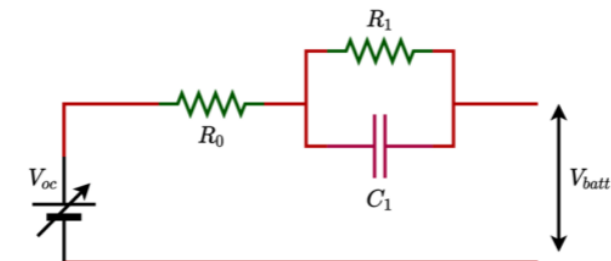
- Gain practical understanding of battery modeling concepts;
- Estimation the battery SoC using end and start point SoC data;
- Model a torque estimator of an e-bike based on measured data from real driving scenarios;
- Test and validate physics-based and data-driven modeling approaches.

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Hybrid Model:
Refined accuracy
&
physics laws still obeyed



CS2: Junction Temperature (T_j) Estimation Model of a 3-Phase Inverter using Deep Learning Models

➤ Task description:

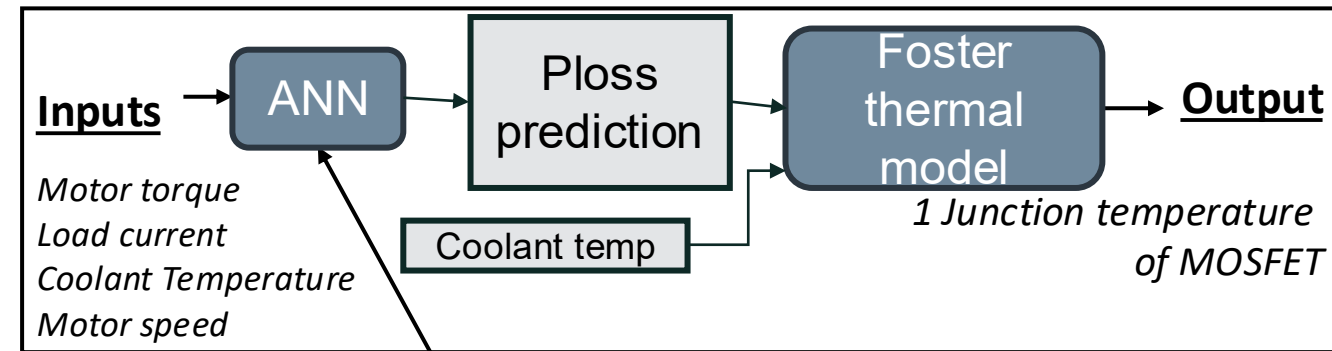
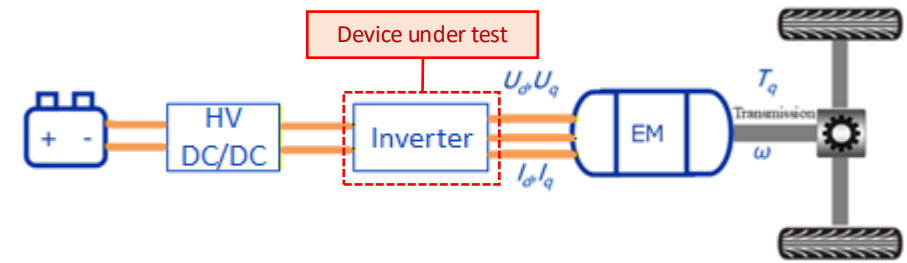
- Estimate the T_j of a MOSFET of a Three-phase Inverter, using deep learning model.
- Data: 16 datasets are given, originating from different city driving cycles. The inputs and output are mentioned on the figure in this slide.
- The problem is popularly known as a Time-Series Regression problem in the machine learning world. Train a deep learning model (ANN, RNN) on the power loss
- Using the predicted power loss, predict the junction temperature with a Cauer or Foster thermal model
- Compare the performance of the different models.

➤ Aim of this case study:

- Learning about time-series regression/prediction modelling using neural networks
- Understanding ANN, RNN and thermal modelling
- Testing of machine learning and physical models on untrained datasets

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Deep learning model

CS3: Classification of Normal and Abnormal Voltage Signals of Power Grid using Unsupervised ML Model

➤ Task description:

- The goal of the work is to detect normal voltage from abnormal voltage waveforms.
- Define the normal and abnormal voltage data extract required features for data-driven modelling.
- Develop an anomaly detector only using normal voltage waveform dataset and test abnormal waveforms with the detector.

➤ Aim of this case study:

- Feature extraction on time-domain data
- **Apply only unsupervised ML models.** Few things to explore are: **Autoencoder, Isolation Forest, One-Class SVM, PCA Reconstruction Error**

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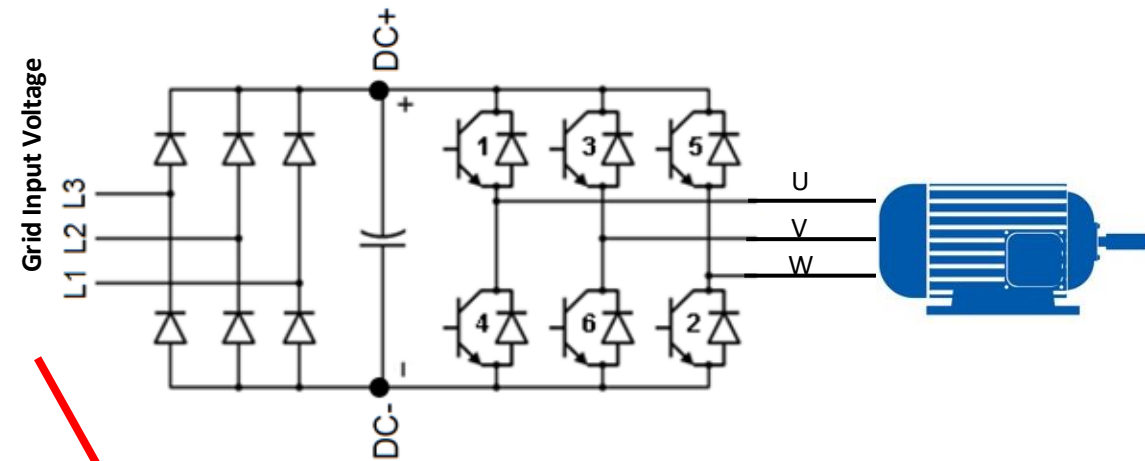
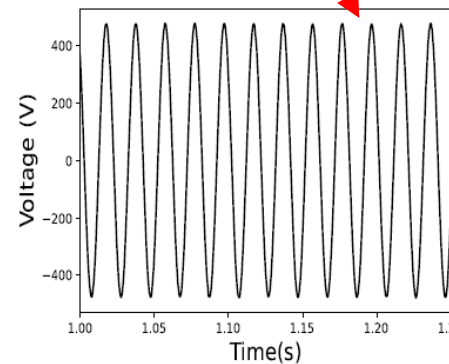
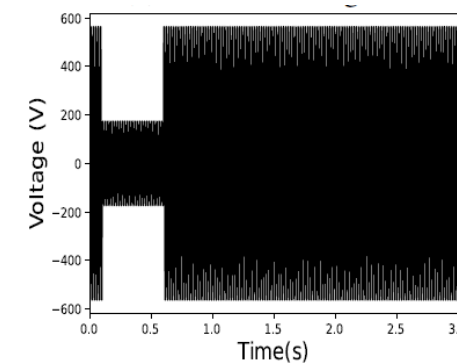


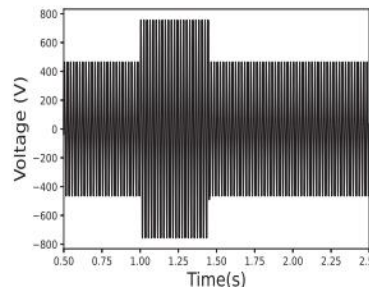
Fig . Typical industrial power converter motor drive



Normal Voltage



Abnormal Voltage



CS4: Parameter Identification of Low-Fidelity Inverter Model for a PMSM Motor Drive

➤ Task description:

- The goal is to design and estimate the parameters of a low fidelity model of 3-phase inverter.
- A rich dataset from testbed of the battery input – 3-phase inverter with a PMSM motor is given.
- The data should be used for extraction and tuning of parameters of the low-fidelity model.

➤ Aim of this case study:

- Data processing of the raw time-series data.
- Use of expert domain knowledge with iterative algorithm to tune the 3-phase inverter model.

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Motor specs: 28 pole pair PMSM motor